Network Vulnerability Assessment using Nmap

1. Introduction

This project focuses on performing network scanning and vulnerability assessment using Nmap. The objective is to identify active hosts, open ports, running services, and potential security risks in a controlled lab environment.

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2. Objective

- To perform network reconnaissance
- To identify open ports and services
- To detect potential vulnerabilities
- To suggest security improvements

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3. Tools Used

- Nmap
- Kali Linux
- VirtualBox (Lab Environment)

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4. Methodology

4.1 Host Discovery

A ping scan was performed to identify active devices in the network.

4.2 Port Scanning

Basic and advanced scans were conducted to identify open ports.

4.3 Service & OS Detection

Service version detection and OS identification were performed.

4.4 Vulnerability Scanning

Nmap NSE scripts were used to detect possible vulnerabilities.

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5. Commands Used

nmap -sn 192.168.137.0/24

nmap 192.168.137.230

nmap -sS -sV -O 192.168.137.230

nmap --script vuln 192.168.137.230

6. Results

The scan was performed on the target IP address:

👉 **192.168.137.230**

The following open ports and services were identified:

- **Port 21 (FTP)** → Open
- **Port 22 (SSH)** → Open
- **Port 80 (HTTP)** → Open

Observations:

- FTP service may allow unauthorized or anonymous access
- SSH service is active and requires secure configuration
- HTTP service is running and may expose web vulnerabilities

7. Risk Analysis

Risk Level	Issue
High	FTP service (possible anonymous access)
Medium	Open ports increase attack surface
Low	Misconfigured services

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8. Recommendations

- Disable unused services (especially FTP)
- Use SFTP instead of FTP

- Implement firewall rules
- Secure SSH (disable root login, use key authentication)
- Regularly update system and services

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9. Conclusion

This project successfully demonstrated how network scanning can be used to identify vulnerabilities and security risks. Proper configuration and security measures are essential to protect systems from potential attacks.

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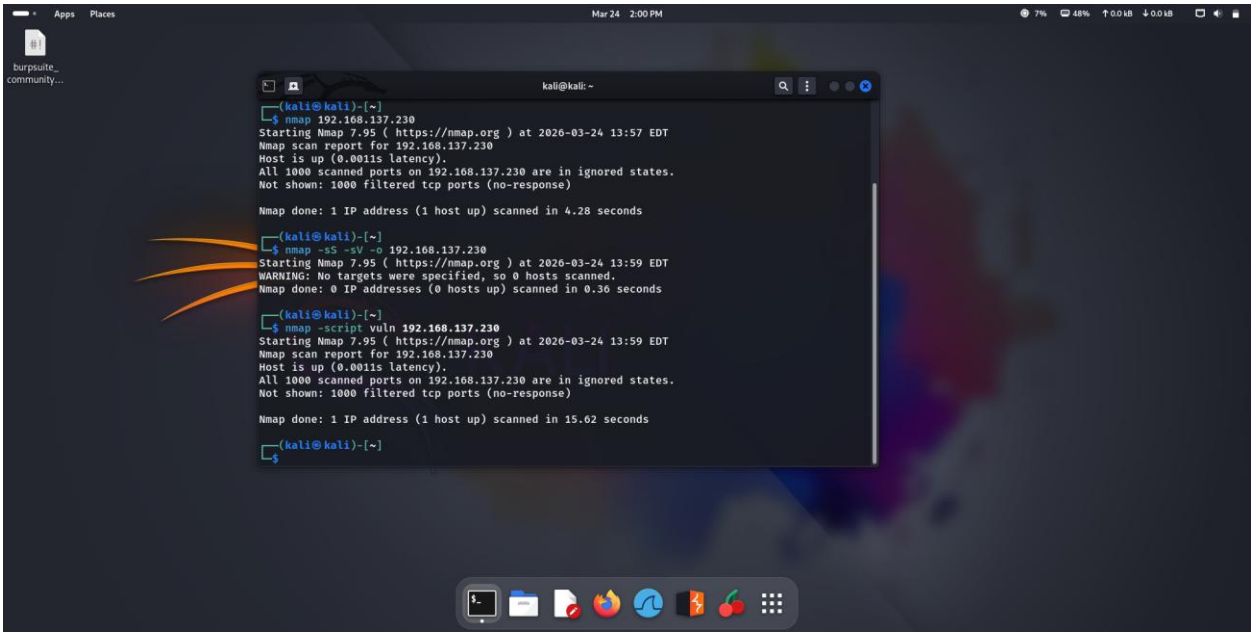
10. What I Learned

- How to perform network scanning using Nmap
- Understanding of open ports and services
- Importance of securing network services
- Hands-on experience in vulnerability assessment


Apps Places
Mar 24 2:00 PM
burpsuite_community...
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.137.230
Starting Nmap 7.95 (https://nmap.org) at 2026-03-24 13:57 EDT
Nmap scan report for 192.168.137.230
Host is up (0.0011s latency).
All 1000 scanned ports on 192.168.137.230 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 4.28 seconds
┌──(kali㉿kali)-[~]
└─$ nmap -sS -sV -o 192.168.137.230
Starting Nmap 7.95 (https://nmap.org) at 2026-03-24 13:59 EDT
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.36 seconds
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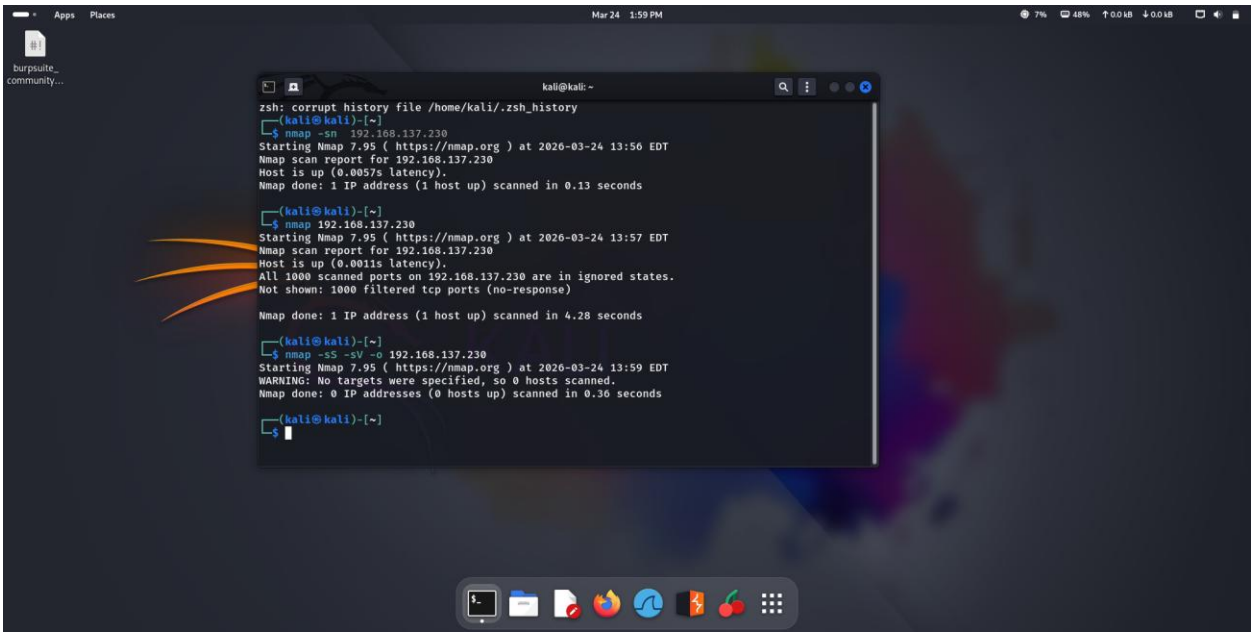
Not shown: 1000 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 15.62 seconds
┌──(kali㉿kali)-[~]
└─$


Apps Places
Mar 24 1:59 PM
burpsuite_community...
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zsh: corrupt history file /home/kali/.zsh_history
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└─$ nmap -sn 192.168.137.230
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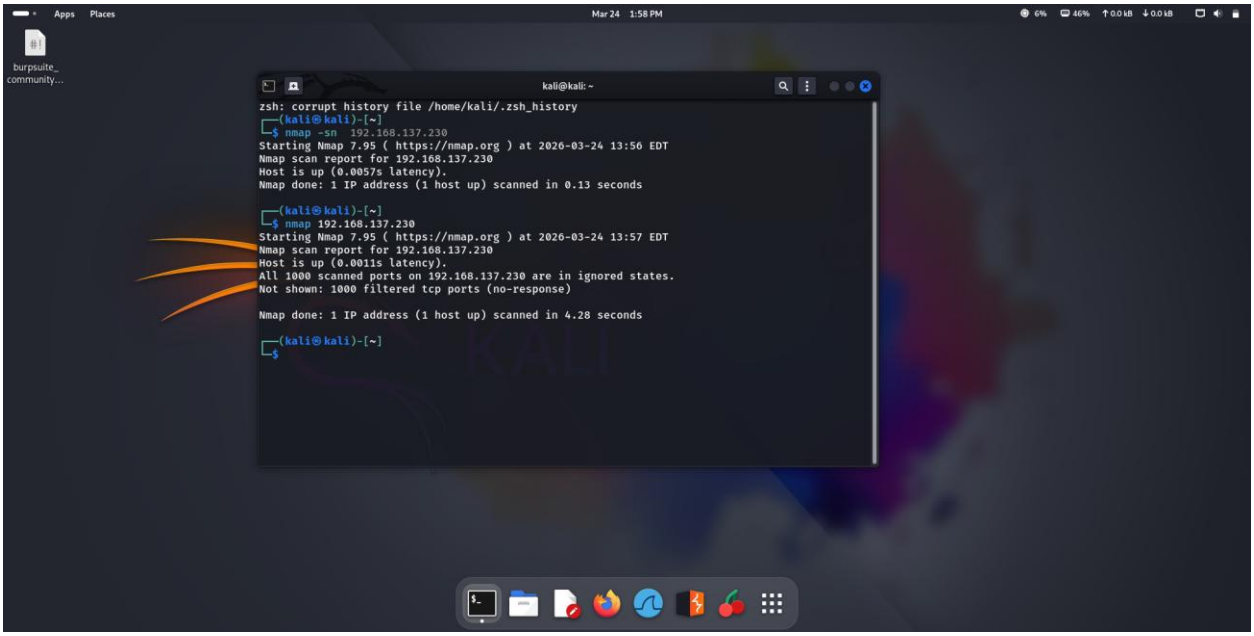
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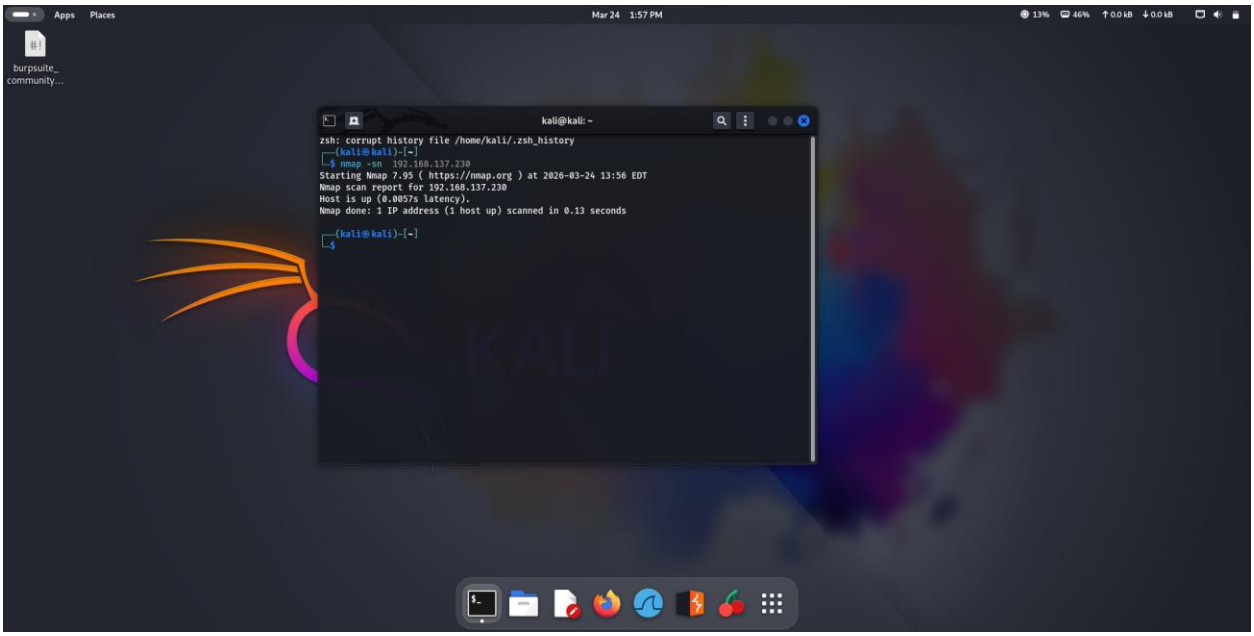
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